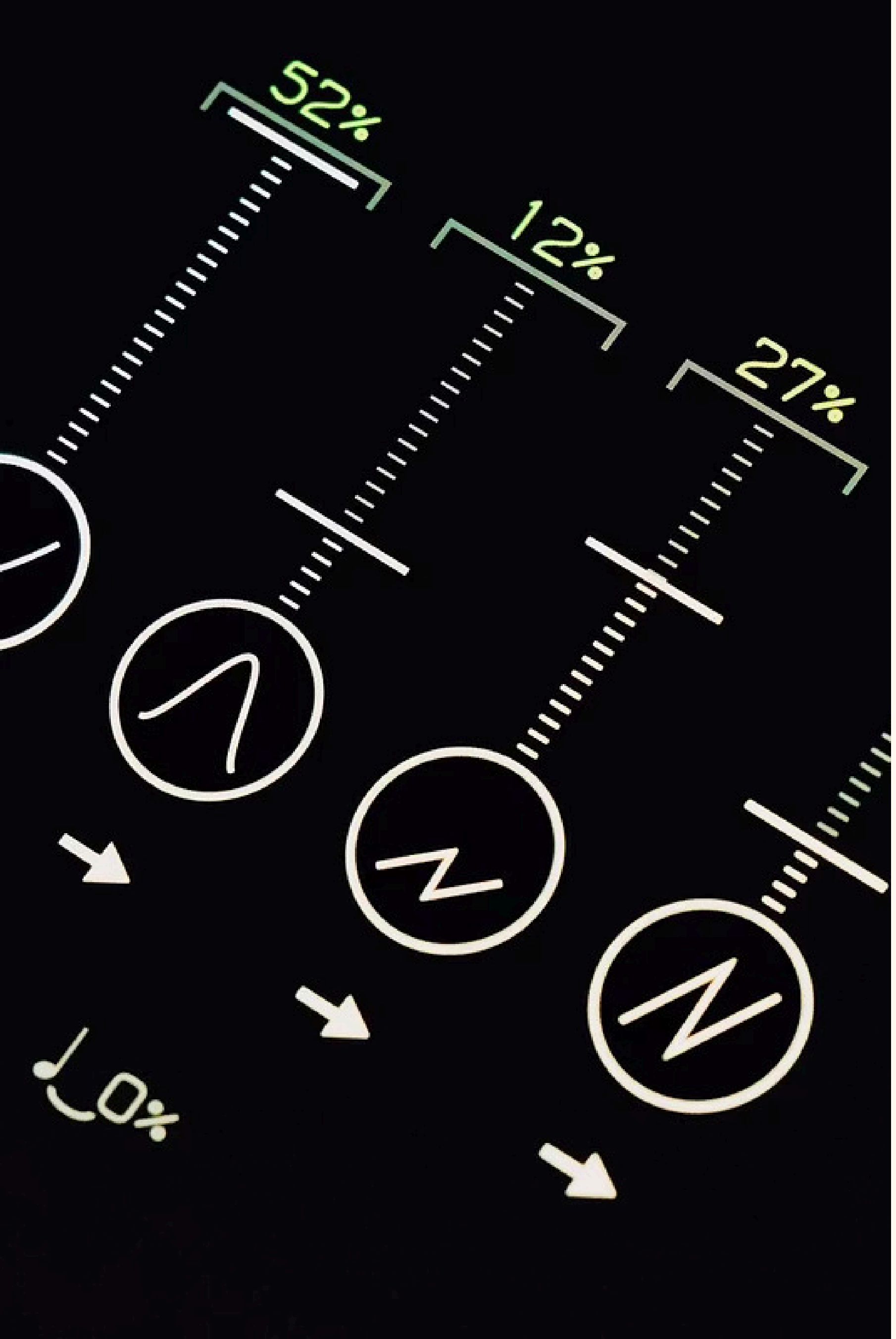


Credit Risk Assessment for Buy-Now-Pay-Later (BNPL)

A Production-Grade ML System for Bati Bank

Presenter: Estifanose Sahilu

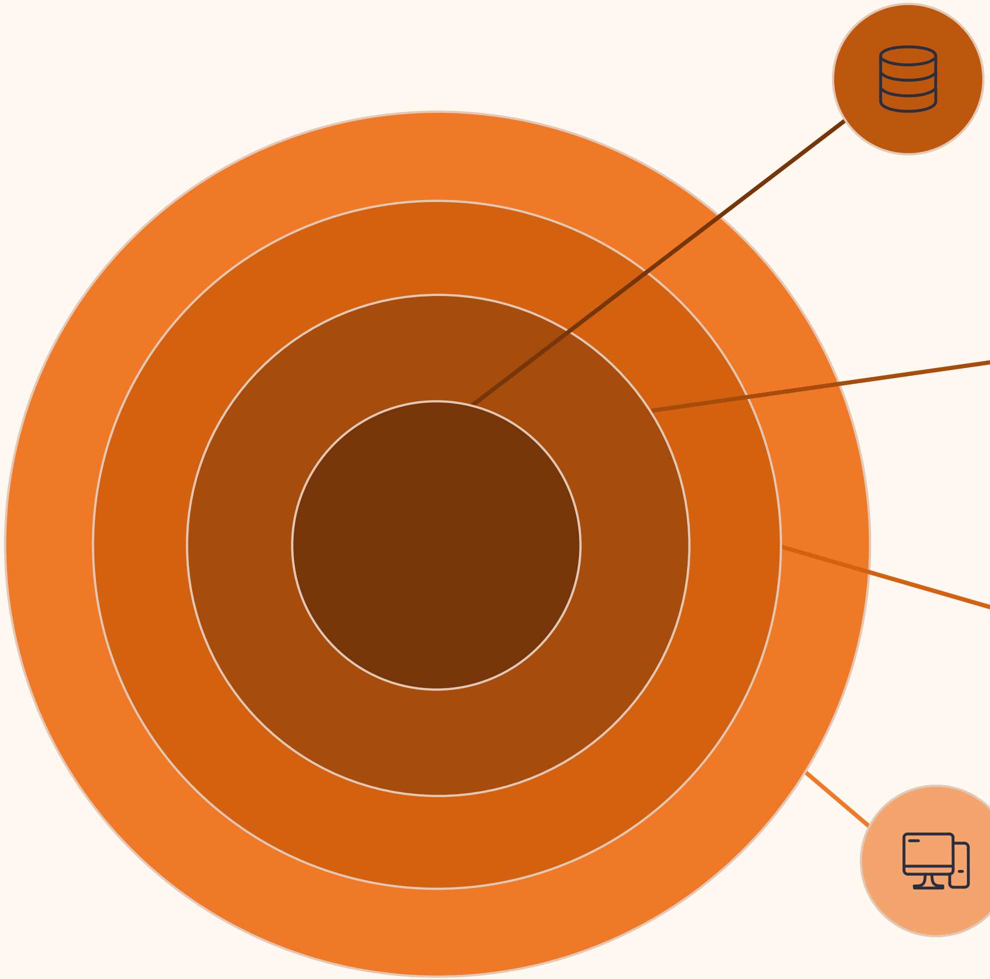
Addressing the "Cold Start" Challenge in BNPL

The Business Imperative

Bati Bank's new Buy-Now-Pay-Later (BNPL) service faced a critical "cold start" problem: an abundance of transaction data, yet no historical default records for new customers. This presented a significant hurdle for accurate credit risk assessment.

Our objective was clear: to minimize default rates while simultaneously expanding credit access to underserved and unbanked populations.

Robust ML Architecture: Powering Real-Time Decisions



Data Layer

Ingest, store, and stream data.

ML Layer

Feature store and model training.

API Layer

Model serving and orchestration.

Frontend

User interfaces and dashboards.

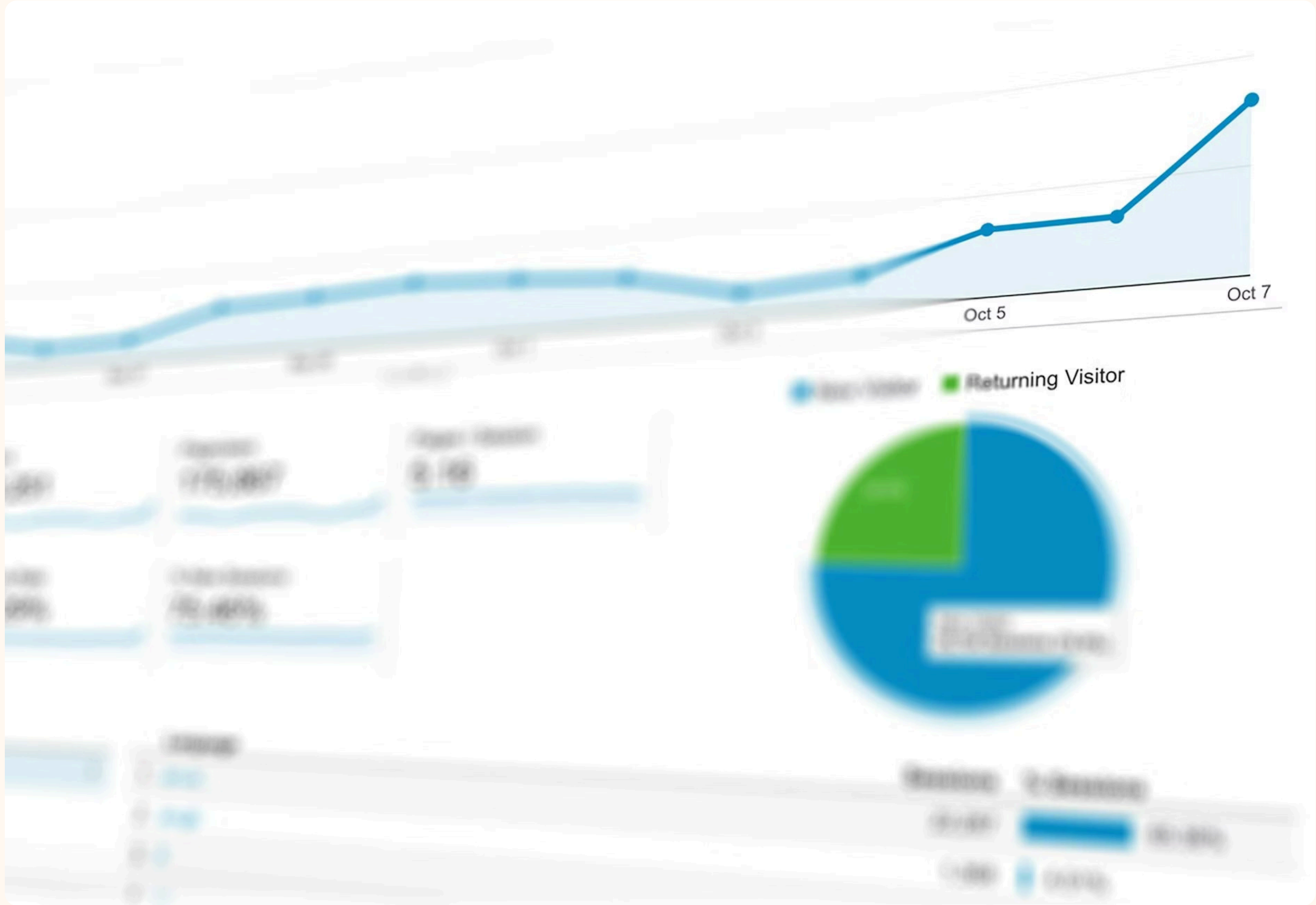
Our system is built on a resilient, modular 4-layer architecture, ensuring scalability and maintainability for dynamic financial services.

Data-Driven Insights: Feature Engineering & Proxy Modeling

From Raw Data to Predictive Power

In the absence of direct default history, we leveraged advanced feature engineering techniques to create a "Proxy Target" for credit risk. **RFM (Recency, Frequency, Monetary) Clustering** was instrumental in segmenting customer behavior and inferring risk profiles.

Our process involved engineering 23 distinct features, including sophisticated WoE (Weight of Evidence) transformations and granular temporal behavior analysis, to capture nuanced risk indicators.



Technical Rigor: Ensuring Reliability and Compliance

Automated Testing

134 automated tests with a 100% pass rate ensure code quality and functional correctness across the entire system.

API Coverage

100% API endpoint coverage guarantees comprehensive testing of all service interactions and data flows.

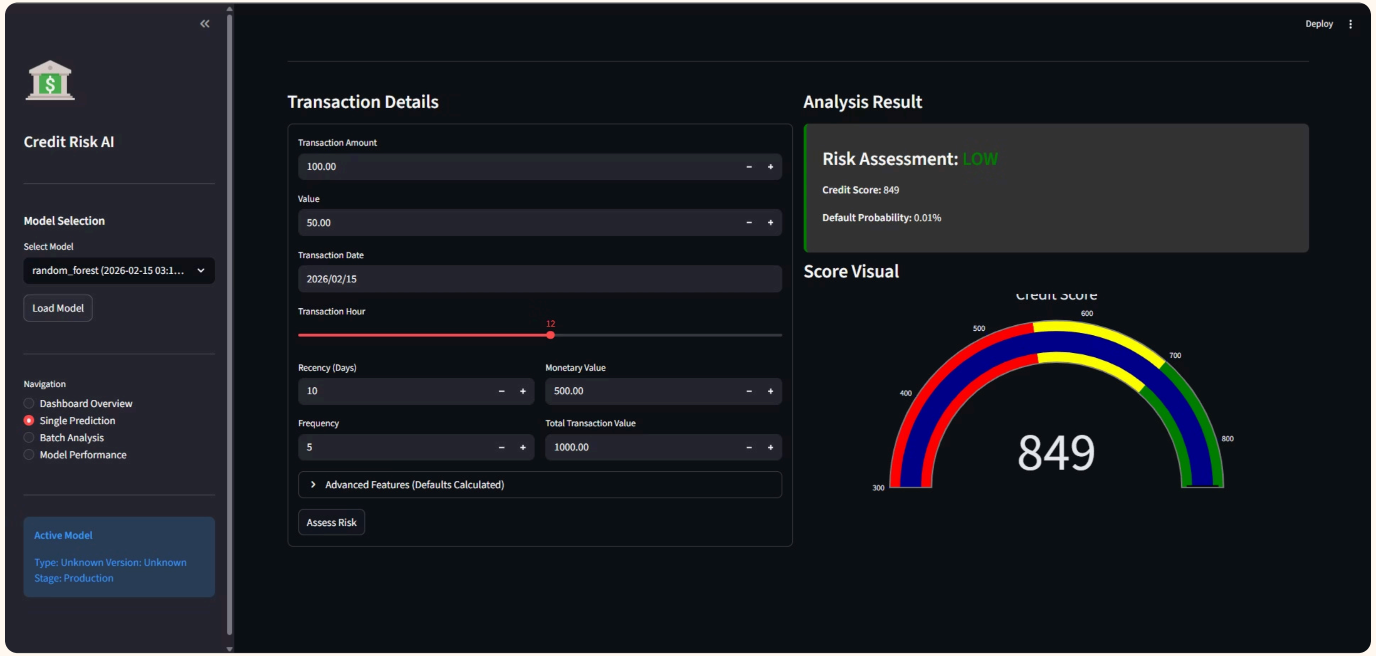
Full Dockerization

Seamless and consistent deployment across environments is achieved through **full Dockerization** of all components.

Basel II Compliance

Focus on **model interpretability and validation** aligns with Basel II regulatory standards for financial risk management.

Model Precision & Interpretability

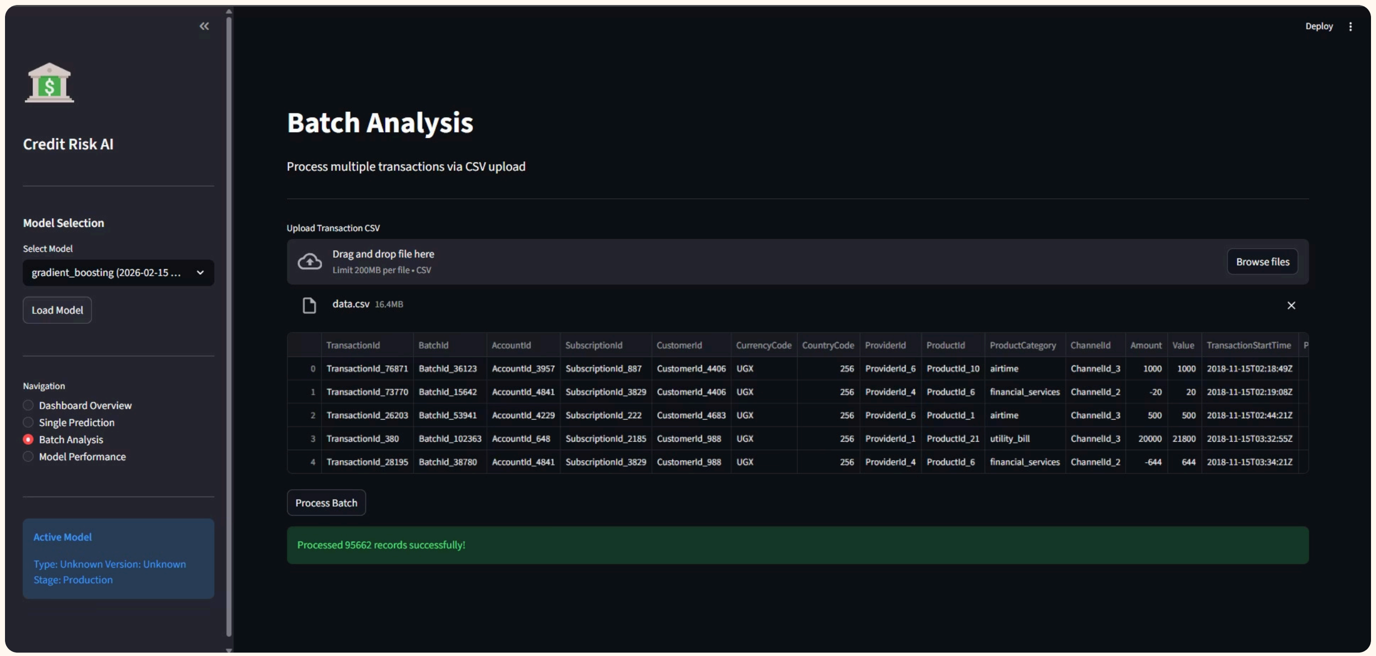


Exceptional Performance & Key Drivers

Our machine learning models achieved an outstanding 1.0 ROC-AUC score on processed features, indicating robust predictive power in distinguishing between high and low credit risk profiles.

Crucial insights revealed that **Monetary value and Frequency of transactions** emerged as the most significant predictors of creditworthiness, underscoring the effectiveness of our RFM-based feature engineering.

Empowering Business Stakeholders: The Streamlit Dashboard



Intuitive Interface for Decisioning

The interactive **Streamlit Dashboard** serves as the primary interface for business users, offering powerful functionalities:

- **Single Prediction:** Real-time scoring for individual customer applications.
- **Batch Analysis:** Efficient processing of CSV files for portfolio-level risk assessment.
- **Model Performance Monitoring:** Continuous oversight of model accuracy and drift.

Tangible Value Proposition & Return on Investment

Risk Mitigation

Automated decline of high-risk profiles based on data-driven insights, significantly reducing potential defaults.

Operational Efficiency

Instant credit scoring (300-850 range) replaces time-consuming manual reviews, streamlining operations.

Scalability

A robust REST API facilitates seamless integration with Bati Bank's mobile and web platforms, supporting future growth.

Conclusion & Future Roadmap

Key Achievements

- Deployed a production-grade ML system.
- Mitigated "cold start" credit risk for BNPL.
- Achieved high predictive accuracy with interpretability.
- Provided a robust tool for business users.

Next Steps & Enhancements

- Real-Time Credit Bureau Integration: Enhance risk assessment with external data sources.
- SHAP Explainability: Implement advanced model interpretability for regulatory and business insights.
- Behavioral Sequence Modeling: Explore deep learning techniques for nuanced user behavior analysis.

Thank You

Your interest in production-grade machine learning for financial services is highly valued.

Questions? Reach out at [My Email] or connect via [[LinkedIn Profile]].